

Paul Lea

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EXPERIENCE

ARGONNE NATIONAL LABORATORY SULI PROGRAM | INTERN

May 2026 - August 2026 | Chicago, IL

- Modeled relativistic electron beams to evaluate electric field-driven phase changes in linear electro-optic mediums.
- Working with RPI's Gaerttner Linear Accelerator Center (LINAC) to facilitate experiment setup and execution.

THZ WAVE ELECTROLYTE SPECTROSCOPY | RESEARCH ASSISTANT

January 2026 - Current | RPI, Troy NY

- Investigated and began to characterize the absorbance properties of Li-Ion battery electrolytes in the THz wave regime.
- Worked with school and department administrators to ensure compliance with campus laboratory policy.

ELECTRO-OPTIC BEAM DIAGNOSTICS | UNDERGRADUATE RESEARCHER

September 2025 - December 2025 | RPI, Troy NY

- Modeled relativistic electron beams to evaluate electric field-driven phase changes in linear electro-optic mediums.
- Working with RPI's Gaerttner Linear Accelerator Center (LINAC) to facilitate experiment setup and execution.

IBM QUANTUM HARDWARE OPTIMZATION |

UNDERGRADUATE RESEARCHER

June 2024 - December 2024 | RPI, Troy NY

- Researched novel approaches to increase relaxation times of IBM Eagle R3 Quantum Processing Units
- Learned Qiskit Pulse control language for QPUs and implemented a containerized python kernel to execute experiments remotely.

PROJECTS

PHYSICS NOTES | ACCESSIBLE ONLINE PHYSICS LEARNING RESOURCE

Fall 2023 - Current | Troy, NY, London, UK

- Currently working to publish a repository of physics resources and notes online, closely mirroring the RPI Physics/Applied Physics curriculum.
- Developed with Quartz, Github Pages and Obsidian

HIGH PERFORMANCE COMPUTING BOOTCAMP | STUDENT

June 2023 - August 2024 | Remote

- Studied techniques and methods to apply high performance computing techniques to problems in physics
- Worked with the Pittsburgh Super-computing Center to practice learned skills in a hands-on environment

EDUCATION

RENSSELEAR POLYTECHNIC IN- STITUTE

MASTERS OF SCIENCE IN PHYSICS

Expected December 2026 | Troy, NY

Cumulative GPA: 4.0 / 4.0

BACHELORS OF SCIENCE IN PHYSICS
AND ECONOMICS

December 2025 | Troy, NY

Dean's List Spring 2024, Summer 2024,
Fall 2024

Cumulative GPA: 3.49 / 4.0

SKILLS

PROGRAMMING

Working Experience:

Python • HTML • Bash

Familiarity:

Fortran • Matlab • Mathematica •

TECHNOLOGY

Linux • Windows • Docker

Nextcloud • Github/Git

MATHEMATICS AND PHYSICS

Computational Physics, Experimental
Physics, Latex

COURSEWORK

GRADUATE

Physics of Advanced Materials

Electrodynamics

UNDERGRADUATE

Introductory Quantum Mechanics

Electromagnetic Theory

Statistical Mechanics

Computational Physics

SOCIETIES

American Physical Society (APS)

Sigma Pi Sigma

Society of Physics Students

Rensselaer Society of Engineering

Innovation (RSEi)

Chi Phi Fraternity

Diversity Equity and Inclusion Physics
Taskforce

LINKS

Website://paulsphysicsnotes.org

Instagram://paulrleaa04

GitHub://paulrlea